

EECE 5554 Robot Sensing and Navigation

Laboratory 05 - Report

Fabricio Valdez Ordonez

Northeastern University

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1 Introduction

In this laboratory exercise, I explored the integration of GNSS and IMU sensing for vehicle navigation, with a focus on magnetometer calibration, inertial heading estimation, velocity integration, and trajectory reconstruction. This report summarizes each step of the analysis and presents comparisons between IMU-based results and GPS-derived ground truth.

2 Questions

This section shows the answer for each question posted on the laboratory git repository.

2.1 How did you calibrate the magnetometer from the data you collected?

What were the sources of distortion present, and how do you know?

To calibrate the magnetometer, I used the data collected while driving in small circles to estimate hard-iron and soft-iron distortions, similar to the previous laboratory. First, I plotted the raw magnetometer measurements in the East–North (E–N) plane and observed that the points formed a shifted, stretched ellipse rather than a centered circle. I computed the hard-iron bias by taking the midpoint between the maximum and minimum values along each axis and subtracting this bias from the raw data. Then, to correct soft-iron effects, I constructed a diagonal scaling matrix from the half-ranges of each axis so that the corrected data were normalized to the average radius.

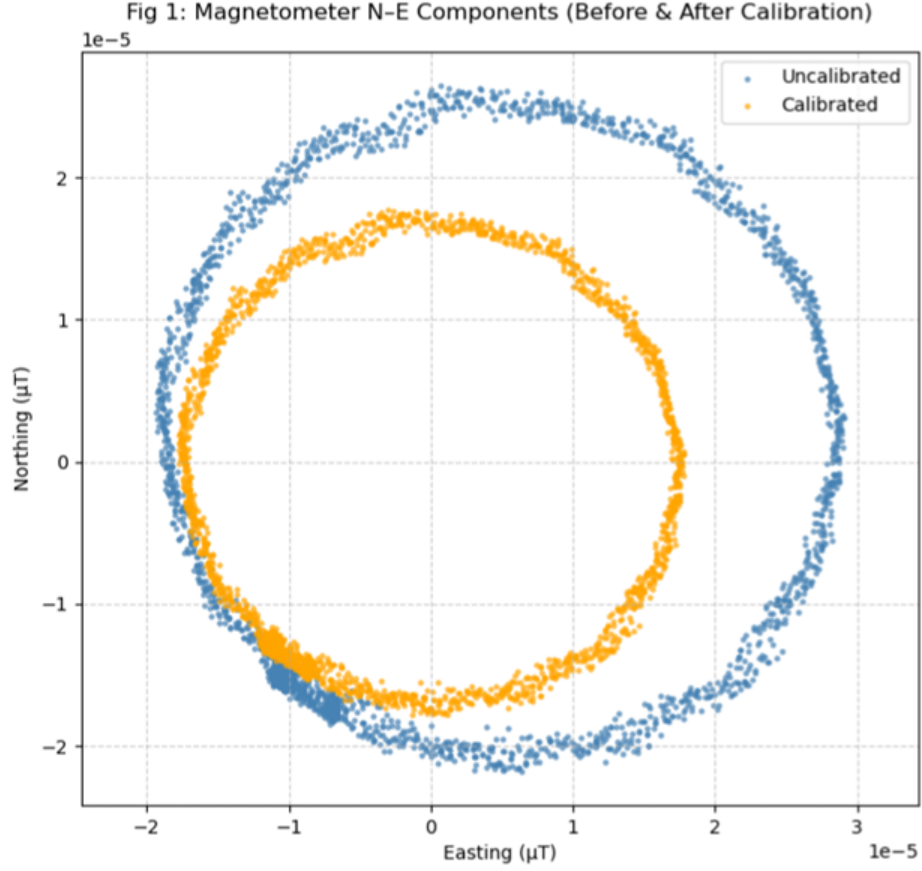


Figure 1: Magnetometer Calibration

The distortions present were due to both hard-iron and soft-iron effects. Hard-iron distortion probably comes from constant magnetic offsets caused by permanent magnets or steady electromagnetic fields from the vehicle's structure and electronics; this appeared in the data as a large offset in all three axes. Soft-iron distortion arises from materials that bend and scale the magnetic field, causing a small elliptical shape in the E–N plot in blue. After calibration, the points became more circular and centered.

After this analysis, I was able to calculate the hard and soft iron distortions that I had to apply to the following analysis for distortion correction. These corrections were represented by the following values:

$$bias = \begin{bmatrix} 4.8550 * 10^{-06} & 2.3550 * 10^{-06} & 4.3755 * 10^{-05} \end{bmatrix} \quad (1)$$

$$soft-iron = \begin{bmatrix} 0.73407698 & 0 & 0 \\ 0 & 0.73712231 & 0 \\ 0 & 0 & 3.55722389 \end{bmatrix} \quad (2)$$

Once we apply this calibration to the data collected, we get the following difference between the magnetometer calibrated and not calibrated.

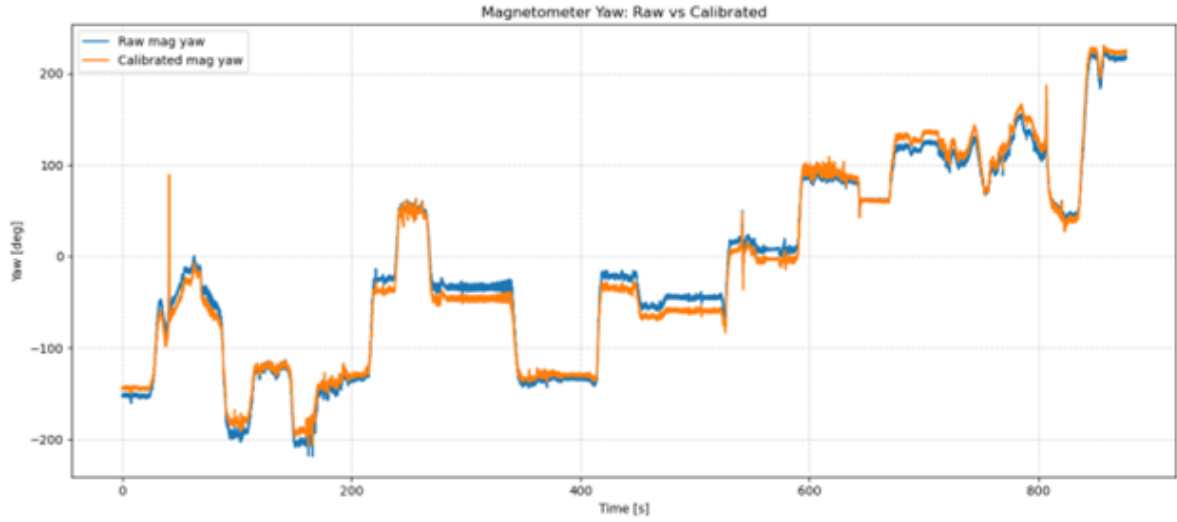


Figure 2: Magnetometer raw vs calibrated

On the other hand, by integrating the gyro, we are also able to identify the yaw behavior. The following figure compares both calculations, gyro-integrated and magnetometer-calibrated.

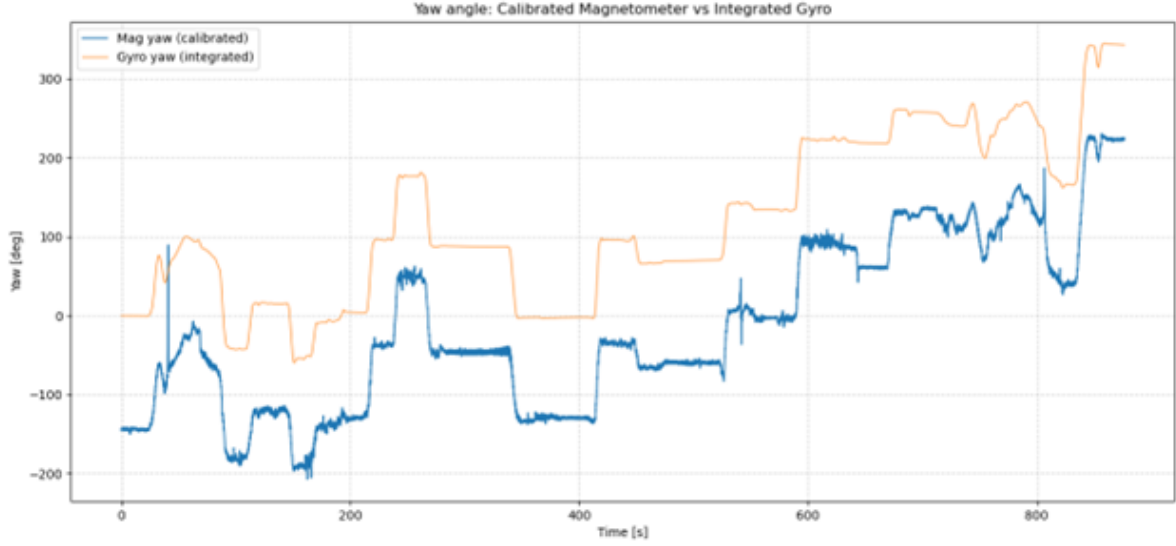


Figure 3: Magnetometer Calibrated vs integrated gyro

2.2 How did you use a complementary filter to develop a combined estimate of yaw? What components of the filter were present, and what cutoff frequency did you use?

To obtain a combined estimate of yaw, I implemented a complementary filter that blends the low-frequency stability of the magnetometer heading with the high-frequency responsiveness of the gyroscope. The gyroscope provides good short-term changes in yaw but drifts over time due to integration error, while the magnetometer provides an absolute heading reference but is noisy and sensitive to magnetic disturbances.

The complementary filter update is represented in the following equation.

$$\psi(k) = \alpha(\psi(k-1) + w_z(k)T_s) + (1 - \alpha)\psi_{mag}(k) \quad (3)$$

We have the previous fused value and the gyro data multiplied by α , combined with the current magnetometer value. The parameter $\alpha = 0.98$ corresponds to a cutoff frequency of approximately 0.5–1 Hz, meaning that fast dynamics come primarily from the gyroscope while slower drift and long-term heading are corrected by the magnetometer. The following figure compares the Low and

high-pass filtered data, the complementary filter output, and the heading estimate of our driving test.

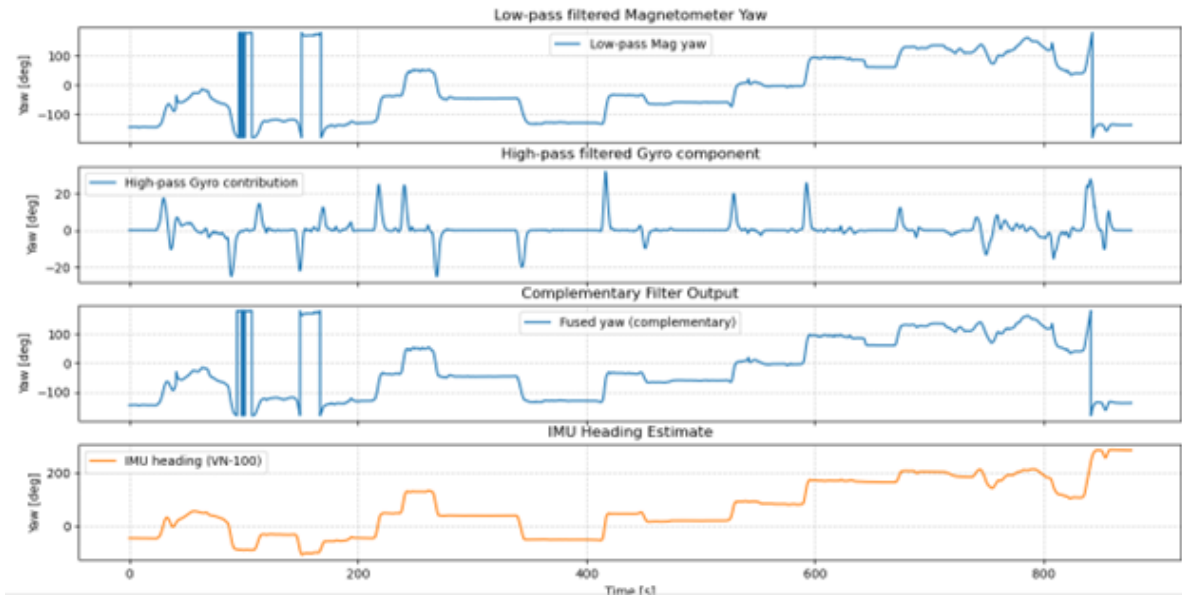


Figure 4: Complementary filter comparison and heading estimate

2.3 Which estimate or estimates for yaw would you trust for navigation?

Why?

I consider that for real navigation, neither the raw gyroscope nor the raw magnetometer yaw is reliable by itself. The gyro yaw drifts over time because even a very small bias accumulates into large heading errors during long driving intervals. The magnetometer yaw, on the other hand, is sensitive to local magnetic disturbances from the vehicle, and nearby electronics, so it produces noise. Because each sensor has a different failure mode, relying entirely on one would cause large navigation errors. Therefore, the most trustworthy estimate for navigation is the complementary-filter fused yaw. With the complementary filter we are able to combine the gyro that provides smooth, short-term heading changes, while the magnetometer provides a long-term reference that prevents drift. By selecting an appropriate cutoff parameter like $\alpha = 0.98$, the filter acts as a stable high-pass on the gyro and a low-pass on the magnetometer, producing a stable heading.

2.4 What adjustments did you make to the forward velocity estimate, and why?

To estimate the forward velocity from IMU data, the first major adjustment was removing gravity from the measured acceleration. Even small roll and pitch angles cause part of the 9.81 m/s^2 gravity vector to project onto the IMU's forward axis, which produces large false accelerations when integrated. Using the quaternion orientation, I rotated the gravity vector into the body frame and subtracted it from the raw accelerometer readings. After removing gravity, I then estimated and removed a constant accelerometer bias, computed from an interval where the vehicle was known to be at rest.

2.5 What discrepancies are present in the velocity estimate between accel and GPS. Why?

In order to talk about the discrepancies, let's first see the plot from the velocity calculated by the integration of the accelerometer, and the plot of the velocity calculated from the GPS data.

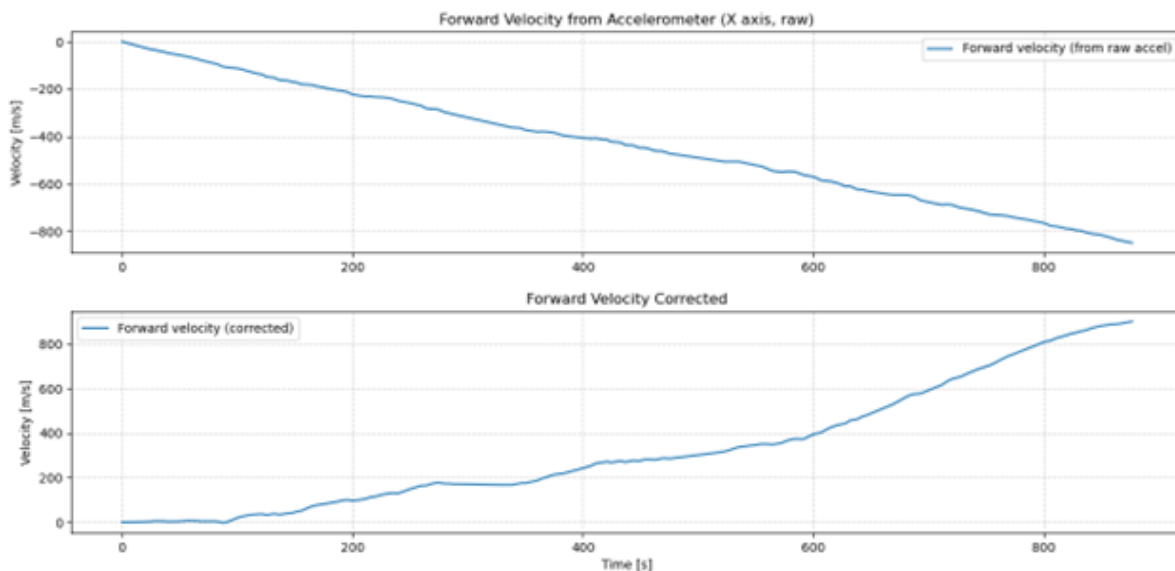


Figure 5: Forward Velocity from Accelerometer

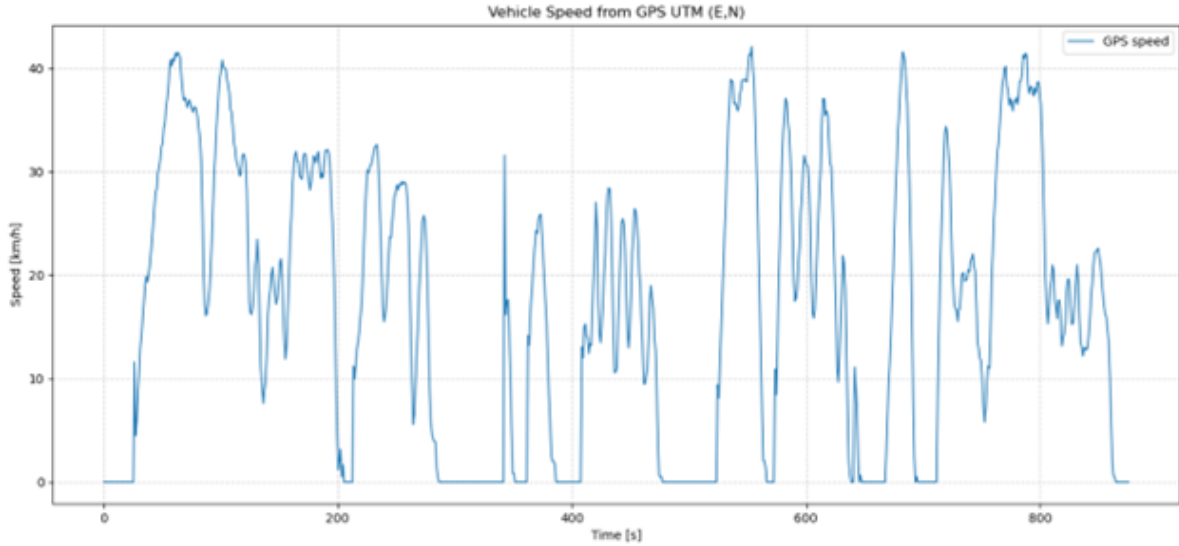


Figure 6: Velocity from GPS data

There is a notable visual difference when comparing the plots. The velocity calculated from the IMU rises up to 800 m/s, even after applying the bias and gravity corrections as described in the previous question; the velocity still keeps increasing. A possible explanation could be that the velocity from IMU is obtained by integrating acceleration, so any other remaining small error in acceleration (gravity leakage, bias, misalignment, or noise) is accumulated over the whole 15-minute drive.

On the other hand, the velocity calculated from GPS seems to be closer to the real velocity experienced during data collection. The range of speed (0 - 40 km/h) is more realistic. At the same time, we have 0 km/h data, which clearly shows stops experienced during the driving test, something missing in the IMU velocity calculated.

2.6 Compute wx' and compare it to y''_{obs} . How well do they agree? If there is a difference, what is it due to?

In order to do this comparison, I started calculating the car motion derivation model first. I started working with the following equation.

$$a_{y,obs} = \frac{d|w_r|}{dt}x_c + 2|w_r|v_y + w_r^2r_t \quad (4)$$

Where I did the following two assumptions:

- The IMU is located in the center of rotation, then $x_c = 0$.
- There is no lateral velocity component $v_y = 0$.
- The radius is proportional to tangential velocity but inversely proportional to w_r .

Then the final modeled acceleration equation was simplified to:

$$a_{y,model} = v * w_r \quad (5)$$

Finally, to compare this lateral acceleration, I decided to compute these calculations with the circle data collected for calibration, as the car is rotating around a referenced center point. The next figure shows the acceleration calculated by this model equation, and the acceleration from the IMU sensor.

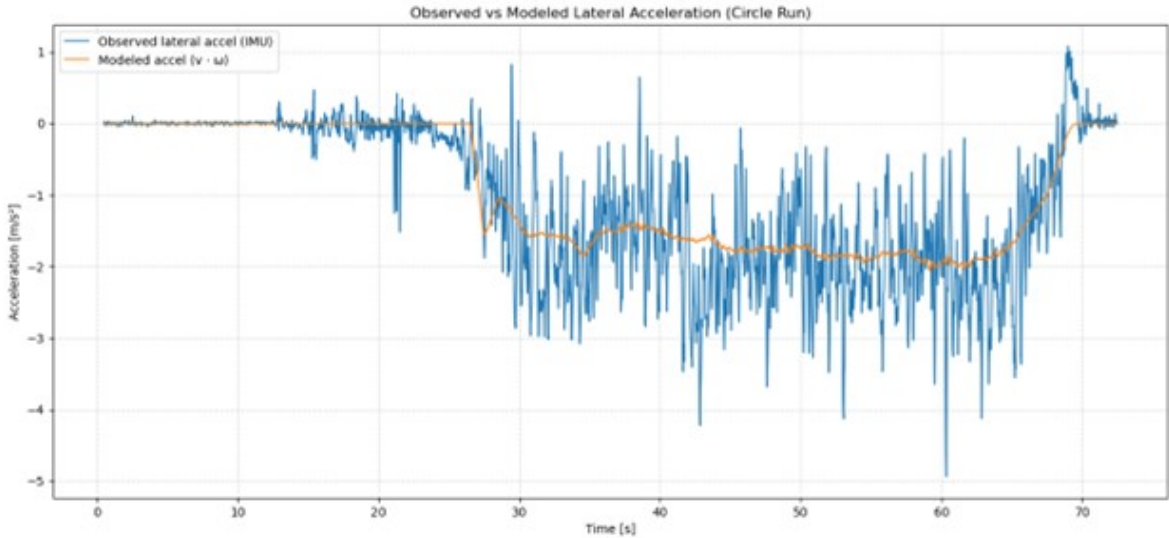


Figure 7: Observed and modeled lateral acceleration

From the figure, we can see that the observed acceleration has higher noise; however, both data sets follow a similar pattern during the test.

2.7 Estimate the trajectory of the vehicle (x_e, x_n) from inertial data and compare with GPS. (adjust heading so that the first straight line from both are oriented in the same direction). Report any scaling factor used for comparing the tracks.

The inertial trajectory has a high scale distance, probably due to a small error that grew exponentially after integration. Therefore, the inertial trajectory was scaled to the GPS trajectory scale. The following figure shows the inertial and GPS trajectories.

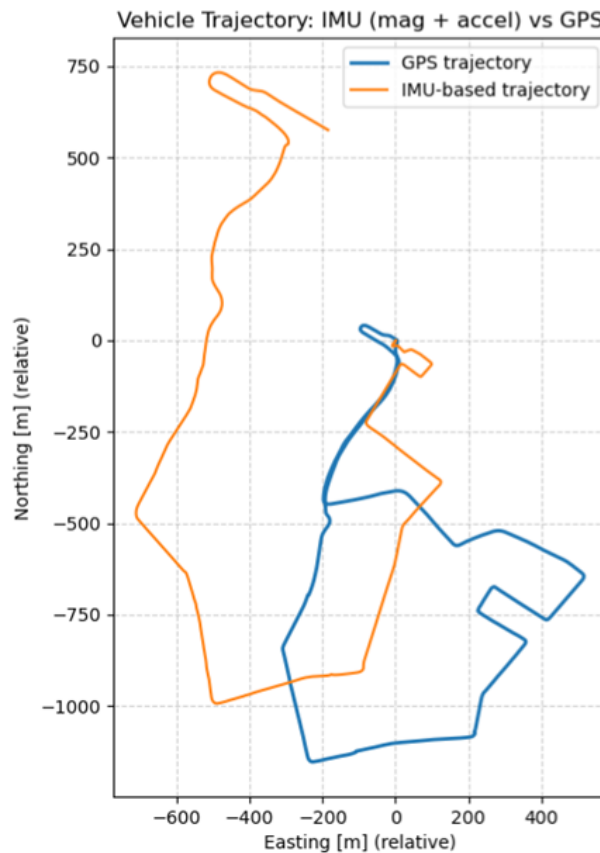


Figure 8: Inertial and GPS trajectories

2.8 For what period of time did your GPS and IMU estimates of position match closely? Given this performance, how long do you think your navigation approach could work without another position fix?

Both estimated trajectories have positions that match closely only at the beginning of the trajectory, probably the first 30 to 60 seconds. My intuition is that there is an inertial drift that was accumulated over time, and this is why both trajectories mismatch. Given these results, I would only trust this navigation approach for the first minute or a maximum of two minutes.

3 Conclusion

This lab showed that using IMU acceleration alone to compute velocity and position produces significant drift over time. Even after gravity and bias corrections, small errors accumulate during integration and quickly cause the IMU-based trajectory to diverge from the true GPS path. In contrast, GPS offers reliable absolute positioning but lacks the smooth short-term response of inertial sensing. Overall, the results highlight that neither sensor is sufficient on its own for navigation.

4 AI usage

During the development of this laboratory, I made use of an AI platform to make English writing and grammar corrections, as well as to learn some Python functions and LaTeX commands.

References

- [1] D. Titterton and J. Weston, *Strapdown Inertial Navigation Technology*. 2nd ed., The Institution of Electrical Engineers, 2004.
- [2] J. A. Farrell, *Aided Navigation: GPS with High Rate Sensors*. McGraw-Hill, 2008.

- [3] J. F. Vasconcelos, G. Elkaim, C. Silvestre, P. Oliveira, and B. Cardeira, “A geometric approach to strapdown magnetometer calibration in sensor frame,” *IEEE Transactions on Aerospace and Electronic Systems*, vol. 47, no. 2, pp. 1293–1306, 2011.
- [4] Q. Li, J. G. Griffiths, and A. G. Dempster, “A fast and robust magnetometer calibration method based on ellipsoid fitting,” *IEEE Sensors Journal*, vol. 17, no. 5, pp. 1383, 2017.